

Feature Engineering and Conformal Prediction in Fish Classification.*

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Description of methods and main findings

1 Introduction

Hydroacoustic signals provide a non-lethal approach for fish classification. However, the features used for classification are high-dimensional and highly correlated. We are also often only given labels for a subset of the data, and must leverage both labelled and non-labelled data to improve classification accuracy. Lastly, we often care to understand which signals are most important for classification, but the correlation among features distorts this signal. The student will use various data science and machine learning approaches to address the data structure challenges and classification tasks.

2 Data

The data set contains **14,575 rows** and 484 columns. Each row represents one fish and it's measurement for a single "ping". A "ping" refers to a time the fish hydroacoustic signal was measured and stored in the dataset. Since multiple signals were recorded for a single fish, there are several rows corresponding the same fish (e.g. there are multiple rows for LT001; Lake Trout fish number one, because multiple measures were taken for this single fish). From now on, we call rows "pings", to familiarize ourselves with how the data was being collected.

The dataset contains information for three fish species: Lake Trout (LT), Lake Whitefish (LW) and Smallmouth Bass (SMB). Table 1 summarizes relevant information about the dataset for each of the species.

*Code and data are available at: <https://github.com/mejiama16/Fish-Project>. The data for this paper was retrieved from the [Fish Tether Experiment GitHub Repository](#).

Table 1: Overview of the dataset

Species	Number of fish	Total pings	Median pings	Min. pings	Max. pings
LT	21	7,248	213	16	1,379
LW	11	4,285	310	33	1,619
SMB	12	3,042	210	14	1,071

The data was obtained from 44 distinct fishes: 21 Lake Trout, 11 Lake Whitefish and 12 Smallmouth Bass. Lake Trout fishes has more pings (instances where data was collected from them), followed by Lake Whitefish and Smallmouth Bass. Although the LW species had less individual fish, more pings were obtained for this species than for SMB. The median pings column shows the median number of pings across all individual fishes for a certain species, in other words, what was the median number of instances where measurements were taken from a single fish. The values are fairly similar across LT and SMB, while LW had more pings in a single fish compared to the other. This is likely because the minimum and maximum number of pings obtained for LW is much higher than for LT and SMB.

It is important to understand the structure of the data and how it is collected, as it largely shapes the statistical method we select and how we make decisions throughout the process. In particular, the rows in this data do not represent single instances or observations, rather, each represents an instance of a subject in time; a measurement taken from a fish. As a result, it is possible for measurements obtained in these instances, although different, may be highly correlated as they come from the same fish. For instance, the rows 1-65 in the raw data set correspond to LT001, Lake Trout fish number one. Because all hydroacoustic signals are taken from this single fish, just at different points in time, it is possible for these to vary very little from measurement to measurement, resulting in highly correlated observations. Additionally, this feature of the data likely violates the assumption of independence of observations, that must hold from the simplest models (OLS regression) to the most complex models in machine learning. We are aware of this violation and consider its implications accordingly.

Now, one might think that the solution is to constraint our data set and analysis to just the 44 individual fish, by randomly selecting one of the measurements taken from each of them. The first issue with this approach is the resulting sample size. We have over 14,000 measurements obtained from this experiment, it seems like exploring the data to overcome the high correlation issue would be a much better solution, as compared to blindly throwing observations away. Another issue with this is: how do we pick which individual pings to keep and which to ignore? Do we hand-pick them? Do we select them randomly? Do we select two per day and ignore the rest? Lastly, when building machine learning models with this data, how can we ensure there is no data leakage? If we keep some observations from fish LT001 in the training set and some in the testing set, it is possible that due to the high correlation between these, the model has already memorized these patterns, leading to an overestimation of the model’s actual capacity.

This paper will touch on some of the decisions we must make when building models using this kind of data, and compare across them. We will do so in the context of dimensionality reduction and feature engineering. In particular, we want to get information regarding which features are the most informative in the classification task. That is, given all the signals we have available for a new, unknown fish, which of these signals are the most likely to help us decide whether this fish is a Lake Trout, a Lake Whitefish or a Smallmouth Bass.

2.1 Time Series Component

Before jumping into the methods it is important to better understand other characteristics of the data. First, the data contains a time series component. The pings for each fish are taken at different times throughout a period of time, no two pings happen simultaneously. Table 2 shows that the data was collected in the span of a week, with SMB having data collected in a much shorter period of time. Note that the data set included other dates, for example `processedDate`, however we keep `sampleDate` to reflect the time the measurement was taken.

Table 2: Span of the data collection process (pings)

Species	Earliest day a ping was made	Latest day a ping was made
LT	2022-07-21	2022-07-28
LW	2022-07-23	2022-07-28
SMB	2022-07-26	2022-07-28

As mentioned earlier, the data set contains a total of 484 columns, out of which 426 are hydro acoustic signals or frequencies. The frequencies go from F45 to F260, and they increase by 0.5. The other 58 columns contain information such as species, unique identifier for the fish and physical characteristics like sex, air bladder weight, tissue sample, length, among others. It is possible that these measures were taken via more aggressive methods; our analysis focuses on just the hydro acoustic signals, as a form to explore non-lethal methods to deal with species classification and problems adjacent to it.

For the purpose of this paper, we do not focus on the temporal data aspect of this problem, though it is possible to start training models based on this particular attribute of the data set. We leave this aspect to be explored in future research.

2.2 Missing Values

Table 3: Missing frequencies in the dataset

Species	F90-F170	F173-F260	Rows with any missing
LT	2,964	183	3,147
LW	789	0	789
SMB	306	0	306
Total	4,059	183	4,242

As observed in Table 3, our data set has 4,242 rows with missing values, most of which come from the Lake Trout species which is also the species for which the most data was collected. It is possible that some of these values are missing at random; for example, the part of the machine that gathered those frequencies was not working that day. While some imputation methods were considered to replace missing values, given the small number of fish in the sample, the consequences of imputing missing values are unknown, and specially when focusing on dimensionality reduction we want to avoid introducing more noise than there already is to perceive signals clearly. Consequently, we drop any rows that have missing values in the frequencies (4,242 in total).

After removing rows with missing frequencies, the resulting data set contains 10,333 rows. Table 4 shows the resulting data set. The number of LT fish is reduced by half. Based on these results, the initial dimensionality reduction will be performed in a binary classification setting between Lake Trout (LT) and Lake Whitefish (LW), given that they are the classes with the most pings (4,101 and 3,496), compared to Smallmouth Bass (2,736).

Table 4: Overview of the dataset with no missing values

Species	Number of fish	Total pings	Median pings	Min. pings	Max. pings
LT	12	4,101	189.5	16	1,379
LW	8	3,496	312.0	96	1,619
SMB	10	2,736	210.0	25	1,071

3 Methods

In this section we compare several techniques for variable selection and dimensionality reduction. We begin with regularization methods, which fit a model containing all p predictors while constraining the coefficient estimates, shrinking them towards zero. We use the standard Lasso and a recent variant, the Whitening Lasso (WLasso), designed for high-dimensional data with highly correlated predictors. We then consider PCA, random forests and boosting.

3.1 Lasso Regularization

Lasso is a shrinkage method that adds an ℓ_1 penalty to the model’s loss function. When the tuning parameter λ is sufficiently large, some coefficients are set exactly to zero, so Lasso returns a sparse model that performs variable selection and is easier to interpret. Because our response is binary (Lake trout vs. Lake whitefish), we use Lasso with logistic regression. Writing $y_i = 1$ for lake trout, x_i for the vector of standardized frequencies of ping i , and w_i for its weight, the coefficients are estimated by

$$\hat{\beta} = \arg \min_{\beta_0, \beta} -\frac{1}{n} \sum_{i=1}^n w_i \left[y_i (\beta_0 + x_i^\top \beta) - \log (1 + e^{\beta_0 + x_i^\top \beta}) \right] + \lambda \sum_{j=1}^p |\beta_j|.$$

Larger values of λ shrink more coefficients to zero, while $\lambda = 0$ recovers ordinary logistic regression. The intercept is not penalized.

Preprocessing. Each frequency was standardized to mean zero and unit variance, as common practice when using Lasso regularization. The means and standard deviations were computed on the training data only and then applied to the test data, so that no information from the test set would enter the model fitting. Additionally, classes differ in the number of fish, and also each fish differs in its number of pings. To prevent fish with many pings (and the larger class) from dominating the fit, each ping received a weight inversely proportional to the number of pings of its fish and to the number of fish in its class, rescaled to average one. This scheme is known as weighted logistic regression, with it each fish contributes equally within its class, and the two classes contribute equally to each other. This helps prevent the model from simply guessing the majority class to achieve a higher accuracy, instead we penalize mistakes in classifying the lower represented class more to account for the fact that the class is underrepresented.

Data splitting and evaluation. Pings from the same fish are not independent, so all splitting was done by fish rather than by ping. Splitting by ping would place pings of the same individual in both the training and test data, which constitutes data leakage and overestimates performance. After removing pings with missing values in the F90–F260 bands, 20 fish remained (12 lake trout and 8 lake whitefish), out of the original 32. The number of pings per fish varies widely (from 16 to 1,619), so the number of pings in any subset cannot be controlled directly. For this reason we report fish-level and balanced accuracy in addition to ping-level accuracy.

First, 4 fish (2 lake trout and 2 lake whitefish) were randomly set aside as a test set (seed = 16). They were not used for fitting or tuning, and they are evaluated once, at the end. The same test fish are used for every method in this study (Lasso, WLasso, random forest, boosting), which allows a direct comparison of their performance on the same unseen individuals.

The remaining 16 fish (10 lake trout and 6 lake whitefish) form the development set. They were randomly divided into $K = 4$ folds of 4 fish, with species spread as evenly as possible

so that each fold contains both species. The penalty λ was chosen by K -fold cross-validation: for each fold, the model was fitted on the other three folds along a grid of λ values and the held-out fold was scored, and the scores were averaged across folds. Folds were formed from whole fish, so no individual appears in both the fitting and the scoring data. We used the binomial deviance as the cross-validation criterion, because, unlike accuracy, it evaluates the predicted probabilities to gauge model confidence as opposed to accuracy. We considered both the value of λ with the lowest cross-validated error ($\lambda_{\min}$) and the largest value within one standard error of the minimum (λ_{1SE}). The final model was then refitted on all 16 development fish with the chosen λ and evaluated once on the 4 test fish.

The fish-level classification was obtained by averaging the predicted probabilities of lake trout over a fish's pings, with a threshold of 0.5. This is mainly because the fish is the real unit: for every ping, the model outputs a probability that it came from a lake trout. A single fish has hundreds or thousands of pings, so we get hundreds or thousands of probabilities for that fish. To get one answer per fish, we average them, and if the average is above 0.5 we call the fish a lake trout, otherwise a whitefish.

Rationale. A single train/validation/test split would leave very few fish for each purpose, and cross-validation on the development fish makes use of all 16 for both fitting and tuning. We chose $K = 4$ so that every fold contains several fish of both species, while each model is still fitted on 12 of the 16 fish. A fixed test set, identical for all methods, was kept so that the final comparison between methods is made on data that played no role in tuning or model selection.

Limitations. With 20 fish in total, all performance estimates are uncertain. The test set contains only 4 fish, so fish-level accuracy can only take the values 0, 25, 50, 75 or 100%, and the number of pings differs greatly between its two species. Test results should therefore be interpreted together with the cross-validated results, not on their own.

4 Results

5 Discussion

6 Appendix